

Customer Shopping Behavior Analysis

1. Project Overview

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

2. Dataset Summary

- Rows: 3,900
- Columns: 18
- Key Features:
 - Customer demographics (Age, Gender, Location, Subscription Status)
 - Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color)
 - Shopping behavior (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)
- Missing Data: 37 values in Review Rating column

3. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

- Data Loading: Imported the dataset using pandas.
- Initial Exploration: Used `df.info()` to check structure and `.describe()` for summary statistics.

	Customer ID	Age	Gender	...	Previous Purchases	Payment Method	Frequency of Purchases
count	3900.000000	3900.000000	3900	...	3900.000000	3900	3900
unique	NaN	NaN	2	...	NaN	6	7
top	NaN	NaN	Male	...	NaN	PayPal	Every 3 Months
freq	NaN	NaN	2652	...	NaN	677	584
mean	1950.500000	44.068462	NaN	...	25.351538	NaN	NaN
std	1125.977353	15.207589	NaN	...	14.447125	NaN	NaN
min	1.000000	18.000000	NaN	...	1.000000	NaN	NaN
25%	975.750000	31.000000	NaN	...	13.000000	NaN	NaN
50%	1950.500000	44.000000	NaN	...	25.000000	NaN	NaN
75%	2925.250000	57.000000	NaN	...	38.000000	NaN	NaN
max	3900.000000	70.000000	NaN	...	50.000000	NaN	NaN

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RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Customer ID                           3900 non-null   int64
1   Age                                    3900 non-null   int64
2   Gender                                 3900 non-null   str
3   Item Purchased                        3900 non-null   str
4   Category                              3900 non-null   str
5   Purchase Amount (USD)                 3900 non-null   int64
6   Location                               3900 non-null   str
7   Size                                   3900 non-null   str
8   Color                                  3900 non-null   str
9   Season                                 3900 non-null   str
10  Review Rating                         3863 non-null   float64
11  Subscription Status                   3900 non-null   str
12  Shipping Type                         3900 non-null   str
13  Discount Applied                      3900 non-null   str
14  Promo Code Used                       3900 non-null   str
15  Previous Purchases                    3900 non-null   int64
16  Payment Method                        3900 non-null   str
17  Frequency of Purchases                 3900 non-null   str
dtypes: float64(1), int64(4), str(13)

```

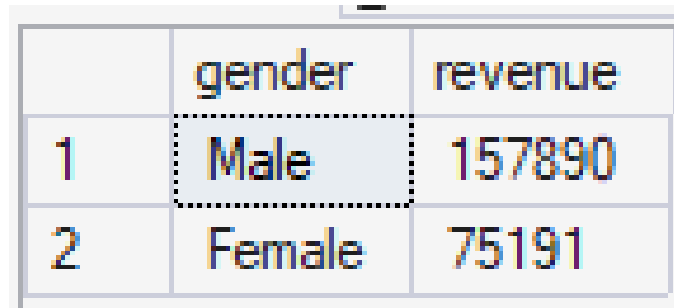
- Missing Data Handling: Checked for null values and imputed missing values in the Review Rating column using the median rating of each product category.
- Column Standardization: Renamed columns to snake case for better readability and documentation.
- Feature Engineering:
 - Created age_group column by binning customer ages.
 - Created purchase_frequency_days column from purchase data.

- Data Consistency Check: Verified if discount_applied and promo_code_used were redundant; dropped promo_code_used.
- Database Integration: Connected Python script to PostgreSQL and loaded the cleaned DataFrame into the database for SQL analysis.

4. Data Analysis using SQL (Business Transactions)

We performed structured analysis in SSMS to answer key business questions:

1. Revenue by Gender – Compared total revenue generated by male vs. female customers.



A screenshot of a SQL query result table. The table has three columns: an index column, a 'gender' column, and a 'revenue' column. The first row is highlighted with a dashed border and contains the values '1', 'Male', and '157890'. The second row contains the values '2', 'Female', and '75191'.

	gender	revenue
1	Male	157890
2	Female	75191

2. High-Spending Discount Users – Identified customers who used discounts but still spent above the average purchase amount.

	customer_id	purchase_amount
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90
10	22	62
11	24	88
12	29	94
13	32	79
14	33	67
15	35	91
16	37	69
17	40	60
18	41	76
19	43	100
20	44	69

3. Top 5 Products by Rating – Found products with the highest average review ratings.

	item_purchased	Avarage Review Ratings
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.8
5	Skirt	3.78

4. Shipping Type Comparison – Compared average purchase amounts between Standard and Express shipping.

	shipping_type	avg_purchased_amount
1	Express	60.48
2	Standard	58.46

5.Subscribers vs. Non-Subscribers – Compared average spend and total revenue across subscription status.

	total_customers	subscription_status	avg_spend	total_revenue
1	1053	Yes	59.49	62645
2	2847	No	59.87	170436

6.Discount-Dependent Products – Identified 5 products with the highest percentage of discounted purchases.

	item_purchased	discount_rate
1	Hat	50
2	Coat	49
3	Sneakers	49
4	Sweater	48
5	Pants	47

7. Customer Segmentation – Classified customers into New, Returning, and Loyal segments based on purchase history.

	customer_segment	total_customers
1	Returning	701
2	Loyal	3116
3	New	83

8. Top 3 Products per Category – Listed the most purchased products within each category.

	item_rank	category	item_purchased	total_orders
1	1	Accessories	Jewelry	171
2	2	Accessories	Sunglasses	161
3	3	Accessories	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

9. Repeat Buyers & Subscriptions – Checked whether customers with >5 purchases are more likely to subscribe.

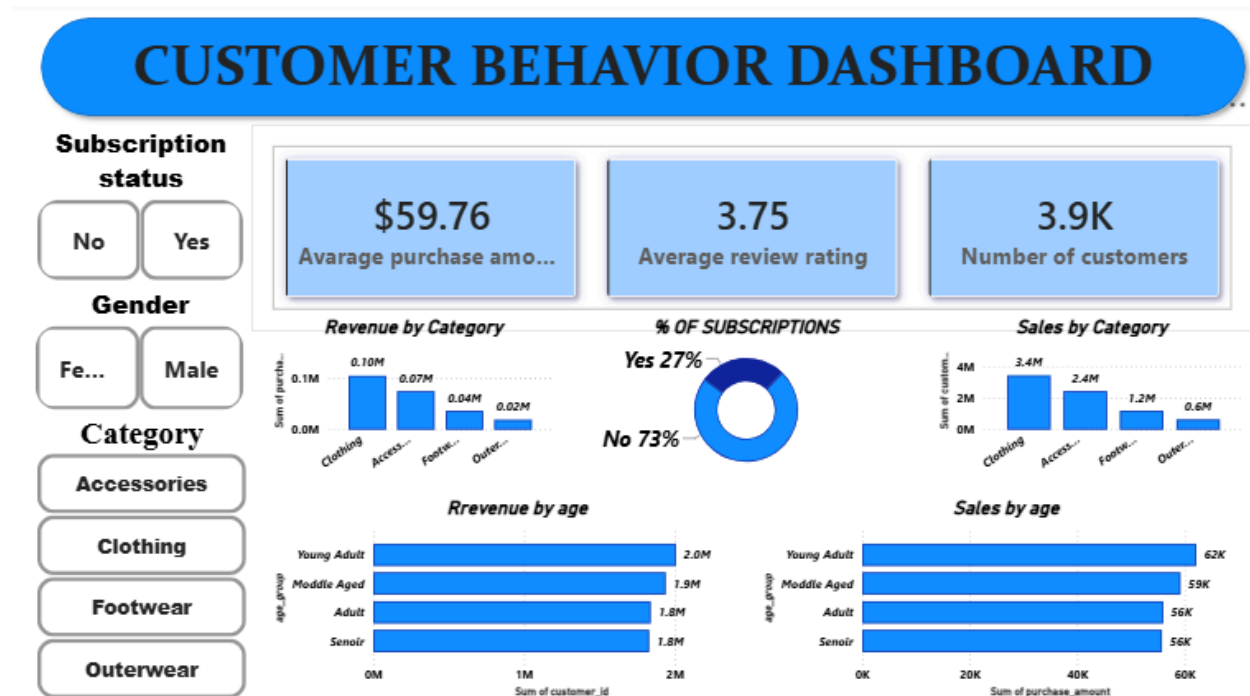
	subscription_status	Repeated_buyers
1	Yes	958
2	No	2518

10. Revenue by Age Group – Calculated total revenue contribution of each age group.

	age_group	total_revenue
1	Young Adult	62143
2	Moddle Aged	59197
3	Adult	55978
4	Senoir	55763

5. Dashboard in Power BI

Finally, we built an interactive dashboard in Power BI to present insights visually.



6. Business Recommendations

- Boost Subscriptions – Promote exclusive benefits for subscribers.
- Customer Loyalty Programs – Reward repeat buyers to move them into the “Loyal” segment.
- Review Discount Policy – Balance sales boosts with margin control.
- Product Positioning – Highlight top-rated and best-selling products in campaigns.
- Targeted Marketing – Focus efforts on high-revenue age groups and express-shipping users.